



A comparison between strain sensing behaviors of carbon black/polypropylene and carbon nanotubes/polypropylene electrically conductive composites

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ABSTRACT

Resistivity variations of carbon black (CB)/polypropylene (PP) and carbon nanotubes (CNTs)/PP conductive composites subjected to tensile strain, that is, the strain sensing behaviors, were studied detailedly. With increasing strain to the maximum value, 3%, the responsivity ($\Delta R/R_0$, R_0 is the original resistance, ΔR is instantaneous change in resistance) increased in exponential and linear fashion for CB/PP and CNTs/PP, respectively, the phenomenon was explained by the tunneling theory. More interestingly, when 10 extension–retraction cycles were applied, for CB/PP, values of max and min $\Delta R/R_0$ increased gradually with increasing the cycle number; while for CNTs/PP, max and min $\Delta R/R_0$ values became more and more negative. It is suggested that the difference mainly depends on the distinction in the microstructure of conductive fillers, the interaction between fillers and matrix and the hysteresis effect resulting from viscoelastic behavior of composites. A schematic illustration was proposed to illustrate the mechanism of the different sensing behaviors.

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1. Introduction

Conductive polymer composites (CPCs) have received great attention due to their popular applications in many fields [1], such as antistatic materials [2], electromagnetic interference shielding [3,4], self-regulated heating materials [5,6], and sensors [7–9]. In recent decades, based on resistivity variation of CPCs exposed to kinds of stimuli: thermal field [10,11], organic liquid [9,12] and external stress field [13,14], etc., external stimuli–response behaviors of CPCs have been investigated more and more frequently. Among them, strain sensing behavior of CPCs has been studied extensively.

Based on the appreciable changes of resistivity in a tensile field, strain sensing behaviors of carbon black (CB) filled CPC have been reported in lots of literature [15,16]. For instance, Fathi et al. [15] investigated the effect of CB structure on electrical and mechanical behaviors of CB/low-density polyethylene (LDPE) and CB/polypropylene (PP) CPCs under the same draw rate. They found that conductivities of all CPCs near optimum CB concentration (defined as the conductive filler concentration at the high end of the percolation region) were sensitive to the applied strain. Specially, for CB/LDPE, the significant strain-sensitivity was observed in composites blended by LDPE and low structure CB with large particle size.

Knite et al. [16] studied the electrical behaviors of CB/polyisoprene CPC in a tensile field; for sample near the percolation transition region, a change of electrical resistance by more than four orders of magnitude was observed when the relative stretch reached 40%, showing a strong responsivity.

On the other hand, based on the admirable characteristics of carbon nanotubes (CNTs), such as exceptional mechanical, thermal, electrical properties and high aspect ratio, the strain sensing behavior of CNTs filled CPC has also attracted many interests in both academic and industrial fields. Pham et al. [17] studied the responsivity of surface resistivity of multi-walled carbon nanotubes (MWNTs)/poly(methyl methacrylate) (PMMA) CPC film in a tensile field detailedly. Their study indicates strain sensitivity of the conductive films can be tailored by controlling CNTs concentration, degree of CNTs dispersion and the film fabrication process. They also found the developed sensors exhibited a broad range of sensitivity. Kang et al. [18] compared the sensitivity and stability of buckypaper with those of the single-walled carbon nanotubes (SWNTs)/PMMA composite under tensile load, they found the strain transfer across the CNTs/PMMA sensor was better than that across the buckypaper owing to the strong polymer interfacial bonding. However, the SWNTs/PMMA strain sensor showed a lower sensitivity than the latter.

Comparison study of strain sensing behaviors between CPCs filled by fillers differing from morphology has also been reported. Das et al. [19] compared the effect of elongational deformation

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on the electrical properties of CB and short carbon fiber (SCF) filled ethylene–propylene–diene rubber (EPDM) and ethylidene norbornene (ENB) (with different ratio) CPCs, it is found the sensitivity against the strain and strain rate depends on both the concentration and the type of conductive filler. SCF filled CPCs, compared with CB filled CPCs, exhibited a higher electrical conductivity at the same level loading. Moreover, electrical resistivity and the mechanical properties of the composites were found to irreversibly alter in extension–retraction cycles. Wichmann et al. [20] studied the electromechanical responses of CB and CNTs filled epoxy CPCs with very low filler contents (0.1 and 0.3 wt.% for the MWCNT composites and 0.5 wt.% for the CB composite), they reported that in the elastic regime, the two composites exhibited a distinct resistance vs. strain behavior due to the different particle geometries. Through applying an analytical model, they pointed out that the strain sensing behavior of CPC may also be used to gain quantitative information about the conducting network, i.e., the number of contacting paths, the interparticle distance, etc.

Based on the reports as mentioned above, it is easy to find that structure of conductive filler, such as the aspect ratio, the particle size, and the microstructure of the aggregates, can cause a remarkable influence on the formation of conductive networks of CPCs, thus make a further influence on the electrical property of CPCs in tensile field. Moreover, although strain sensing behaviors of CPCs has been investigated by many researchers, the mechanism of strain sensing behavior in tensile field is still unclear [20], and further research of CPCs built by conductive filler with different microstructure and geometry is required. Understandably, the comparison study can be used as a probe to reveal the microstructure evolutions of CPCs in tensile field and in favor of the understanding of the sensing mechanism, more importantly, this exploration is useful for the application of the CPC as a strain sensor.

In the present work, CB and CNTs, which are two typical carbon category fillers with different characteristics, especially the aspect ratio, were chosen as conductive fillers. PP, which is a general polymer, was used as the matrix. CB/PP and CNTs/PP CPCs were prepared to construct conductive network with different geometric structure. The morphology and percolation behaviors were studied firstly. In order to compare the responsiveness and reproducibility of two CPCs, strain sensing behaviors of the two CPCs were investigated in detail during 10 extension–retraction cycles with a maximum strain, 3%. The influence of the filler morphology on strain sensing was discussed. The sensing mechanism, which is determined by the microstructure of conductive network and its evolution in the tensile field, was revealed. Based on the discussion, a schematic illustration was proposed to explain the change of the strain sensing behaviors of these two CPCs.

2. Experimental

2.1. Materials

Materials: isotactic PP (T30S) used in this study is a commercial product of Dushanzi Petroleum Chemical Co. (Xinjiang, China). Its M_w is about 3.99×10^5 g/mol with the molecular weight distribution *ca.* 4.6 and the melt flow index (MFI) 3.0 g/10 min (230 °C, 2.16 kg). The CB, model VXC-605, was provided by Cabot with a dibutyl phthalate absorption value of 148 ± 15 cm³/100 g. CNTs were purchased from Chengdu Organic Chemicals, Chinese Academy of Science, with a diameter 20–40 nm and a length 50 μm, they were grown via the chemical vapor deposition method. Both CB and CNTs were dried at 80 °C for 10 h to get rid of the water before use.

Fig. 1 shows the typical morphology of CB and CNTs used in this study. In Fig. 1a, it is found the size of the primary CB particles is

about 30–50 nm, even so, most of CB particles exist as aggregates with branchings, which are favorable for the formation of conductive paths. It is known that CNTs have inherent entanglement, large specific surface area, high surface energy and strong van der Waals forces between nanotubes. In Fig. 1b, it is evident that CNTs show extremely large aspect ratio compared to CB [21]. This characteristic results in poor disperse of CNT in polymer matrix, but it is beneficial to forming of the microstructure of conductive network.

2.2. Fabrication of CPCs

CB/PP and CNTs/PP composites with different filler concentration were prepared by a Haake internal mixer (Haake polylab system-Rhemex 252p series, RC9000) at the mixing temperature 190 °C. The mixing time is 10 min, the rotate speed was set as 50 rpm. The mixture was then pelletized, and compression molded into plates at 200 °C for 5 min with a pressure of *ca.* 15 MPa. Dimension of samples for conductive properties test and strain sensing test is $150 \times 10 \times 0.8$ mm³.

2.3. Test conditions

For strain sensing test, a displacement controlled Suns UTM2203 universal testing machine equipped with a 2000 N load cell was applied. The specimen with a gauge length of 50 mm, while two copper electrodes separated 40 mm were bonded with silver paste to measure the electrical resistance (*R*) with a high resistivity meter. At least five specimens were tested for each experiment of the two CPCs, representative results are presented. The schematic figure for the specimen is displayed in Fig. 2. In order to investigate the evolution trend of stress–strain curve in tensile field, the cross head speed for loading and unloading was set as 0.5, 3 and 5 mm/min, respectively. In order to study the strain sensing reproducibility of CPCs, 10 extension–retraction cycles were conducted at the tensile speed of 0.5 mm/min. During the mechanical test, the variation of the resistance in the tensile process was recorded continuously with a computer coupled with the high resistivity meter (model TH2683, supplied by Changzhou Tonghui Electronics Co., Ltd.).

2.4. Morphological observation

The specimens were fractured after frozen in liquid nitrogen for 1 h. The fracture surfaces were covered with a layer of gold to make them conductive, and then observed with a 7500F JEOL scanning electron microscope.

3. Results and discussion

3.1. Percolation behaviors

In order to investigate the electrical response behaviors of CB/PP and CNTs/PP in tensile field, firstly, the percolation behaviors of both CPCs were studied. Fig. 3 shows the effect of filler content on the volume resistivity of CNTs/PP and CB/PP CPCs. For CB/PP, at a low filler content, the resistivity decreases weakly with increasing the filler content. When the CB content is increased from 6.90 to 7.16 vol.%, a 10 orders of magnitude drop in resistivity, that is the so-called percolation behavior [22], is observed. The percolation threshold is estimated to be *ca.* 7.05 vol.%. This sharp change in resistivity is a result of a high percentage of electrons flow through the specimen [23], which results from the formation of an interconnected structure of CB particles. This percolation threshold is a little higher than reported literature because a relatively high melt flow PP and a low structure CB were applied in the

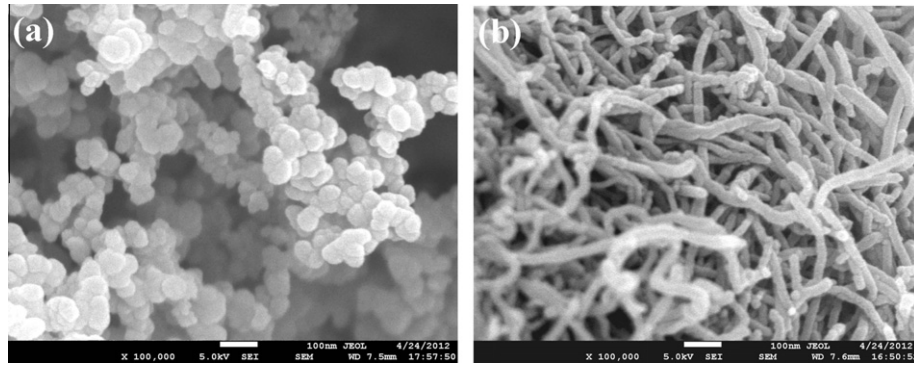


Fig. 1. Scanning electron microscope (SEM) images of CB (a) and CNTs (b) at a high magnification.

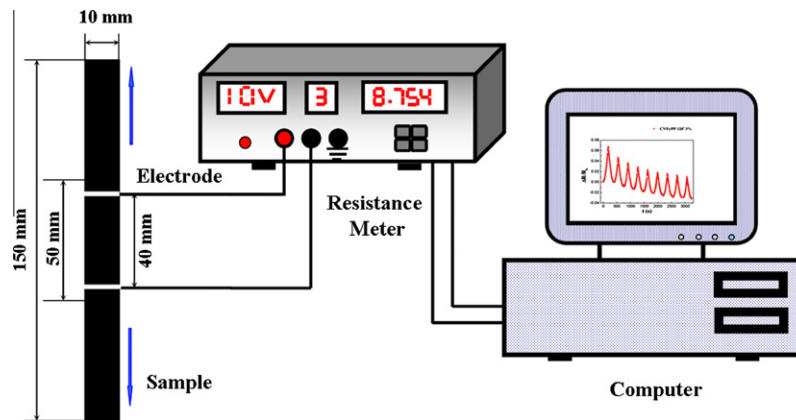


Fig. 2. Schematic diagram of a sample for strain sensing test. Blue arrows represent the direction of stretching. White lines on the sample represent the silver paste. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

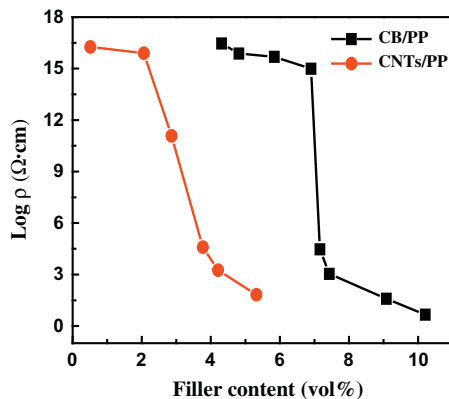


Fig. 3. Volume resistivity vs. CB and CNTs contents of CB/PP and CNTs/PP. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

present paper [24]. For CNTs/PP, increasing the CNTs content from 3.07 to 3.41 vol.%, an 11 orders of magnitude drop can also be observed. The percolation threshold *ca.* 3.24 vol.%, which is much lower than that of CB/PP, is obtained. The lower percolation threshold of the CNTs/PP is most likely related to the geometry of two different fillers (the particular high aspect ratio morphology of CNTs, see Fig. 1). Above the percolation threshold, the resistivity of both CPCs levels off.

It is reported that the sensitivity of resistivity to tensile strain is more remarkable when the CPC contains an optimum filler concentration just beyond the percolation region [15,16]. The reason is

that at this concentration, any distortion of the inter-aggregate distance can generate a change in potential barrier that more easily makes the transport of charge carriers impossible, and as a consequence, a significant change in the resistivity will be produced [25]. In other words, strain sensitivity of resistivity largely depends on the conductive filler concentration [17,19]. In the present paper, the approximate optimum filler concentrations 7.16 vol.% CB and 3.41 vol.% CNTs for the two CPCs are chosen.

The morphology of these two CPCs at the concentrations as mentioned above was observed. Fig. 4 shows the SEM micrographs of the fractured surfaces of the two composites, which show well-dispersed CB and CNTs in PP matrix.

3.2. Stress–strain behaviors under different tensile speeds

The tensile experiment was conducted first with different tensile speeds to determine the mechanical deformation behaviors of CB/PP and CNTs/PP.

Fig. 5 displays the stress (σ)–strain (ε) behaviors of CB/PP (Fig. 5a) and CNTs/PP (Fig. 5b), respectively. Obviously, the stress vs. strain plots show that both composites are brittle in tension, which is the typical mechanical behavior for this kind of CPCs filled with a lot of rigid filler particles. It is also found the shape of the stress–strain curves is similar to each other even at different tensile speeds. In tension process, it is known macromolecular chain segments' motion always cannot match the change rate of the stress, resulting in the creep behavior, hysteresis effect, etc. This phenomenon is affected by the tension speed significantly: if a low tension speed is applied, the macromolecular chain segments have time to relax, showing a high elongation at break and a low

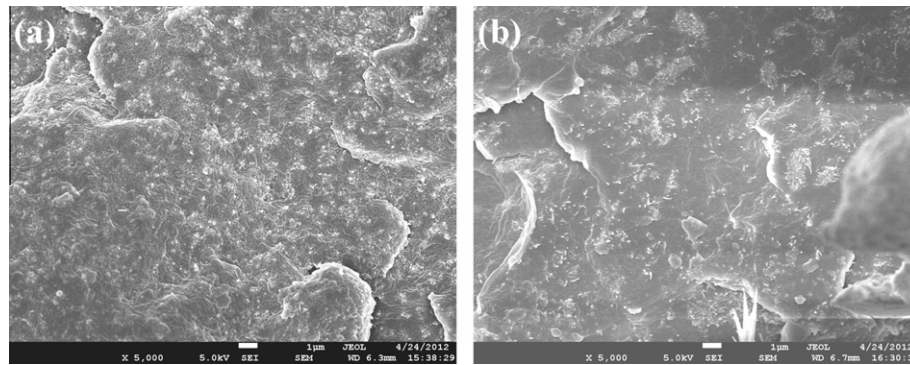


Fig. 4. SEM micrographs for the morphology of (a) CB (7.16 vol.%) / PP and (b) CNTs (3.41 vol.%) / PP CPCs.

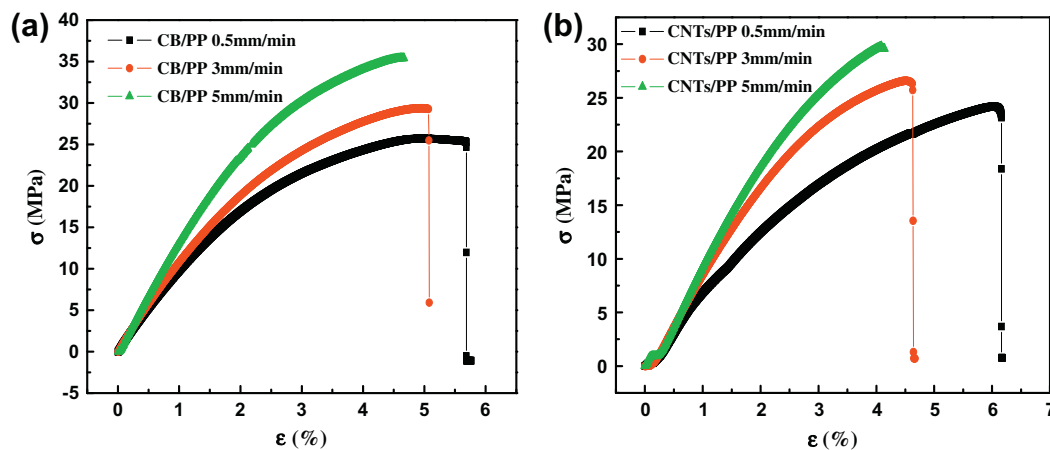


Fig. 5. Stress (σ) as a function of strain (ϵ) at different tensile speeds: (a) CB (7.16 vol.%) / PP and (b) CNTs (3.41 vol.%) / PP composites. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

The stress (σ_b) and strain (ϵ_b) of CB (7.16 vol.%) / PP and CNTs (3.41 vol.%) / PP CPCs at break.

Tensile speed	0.5 mm/min		3 mm/min		5 mm/min	
	σ_b	ϵ_b	σ_b	ϵ_b	σ_b	ϵ_b
CB/PP	25.26	5.68	29.29	5.06	35.28	4.74
CNTs/PP	23.16	6.16	26.32	4.62	29.80	4.09

strength; if the tension speed is increased, the macromolecular chain segments have no enough time to relax, a lower elongation at break and higher strength are thus obtained. In this paper, with increasing the tensile rate, similarly, the stress at break of CB/PP and CNTs/PP both increases, while the strain at break decreases (see Table 1, σ_b and ϵ_b represent the stress and strain at break for the CPCs). In order to capture detailed information about the development of the conductive network and obtain a higher responsiveness during the test, a relatively large deformation (3%) and a low tensile speed 0.5 mm/min are chosen to study the electrical and mechanical responses of CB/PP and CNTs/PP containing different geometrical conductive networks in this paper.

3.3. Mechanical properties under cyclic tension

In order to get information about the influence of conductive filler microstructure and its evolution on the mechanical and electrical responses of the two CPCs, 10 extension–retraction cycles were performed with the maximum strain 3%.

Fig. 6 shows the 10 cycles of stress–strain loading records of the two CPCs. For CB/PP, in each cycle the film stress increases almost linearly with the applied tensile strain up to the maximum strain (see the inset in Fig. 6). After reaching the maximum strain, the machine cross-head moved to original point. During these cycles, the hysteresis effect which results from the deformation and viscoelastic behavior in tensile process, is observed as reported in literature [25–28]. The strain does not restore to the original point when the stress return to 0 [29]. Furthermore, a slight decrease in maximum stress of the two CPCs occurs with increasing the cycle number (see the inset in Fig. 6), which is related to the viscoelastic nature of PP, that is, polymer matrix have no enough time to relax.

3.4. Electrical properties under cyclic tension

In order to explore the relationship of electrical and mechanical properties, the evolution about the stress and relatively electrical resistance $\Delta R/R_0$ as a function of the strain of both CPCs was studied firstly.

Fig. 7 shows the typical stress, $\Delta R/R_0$ –strain curve in tension. It can be seen that the stress increases linearly with the strain for both CPCs. However, for $\Delta R/R_0$ vs. strain curve of CB/PP, $\Delta R/R_0$ increases in a non-linear fashion. For CPCs during elongation, it is known that two phenomena occur simultaneously in the system: breakdown of existing conductive networks and formation of new conductive networks. These two phenomena compete with each other during the whole process [30]. For CB/PP, at a small strain below 0.5%, the number of the formation and breakdown

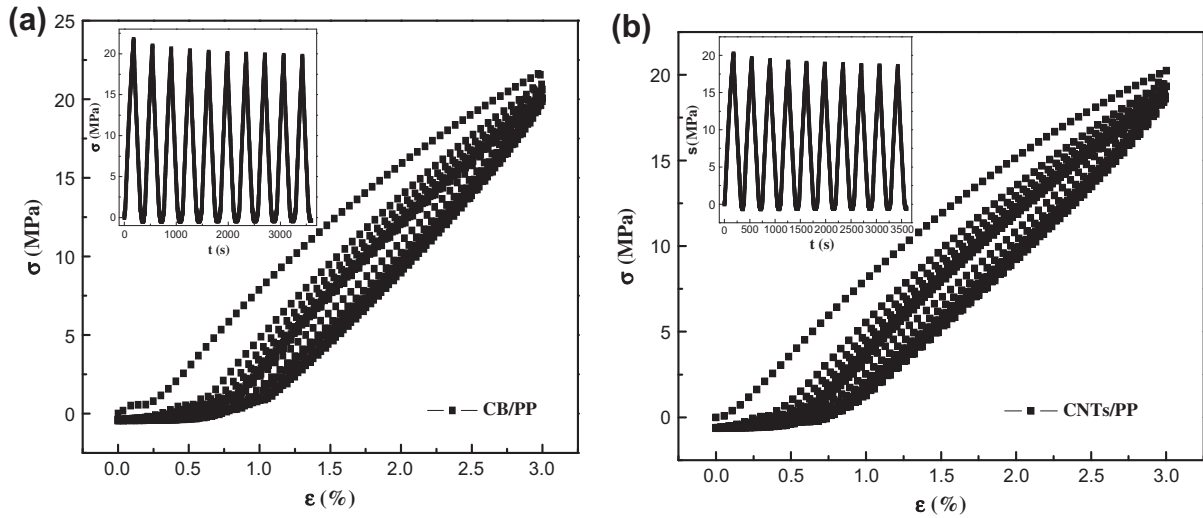


Fig. 6. Relationships of stress vs. strain in (a) CB (7.16 vol.%) / PP and (b) CNTs (3.41 vol.%) / PP composites during 10 extension–retraction cycles. The inset is the plot of stress vs. time (t).

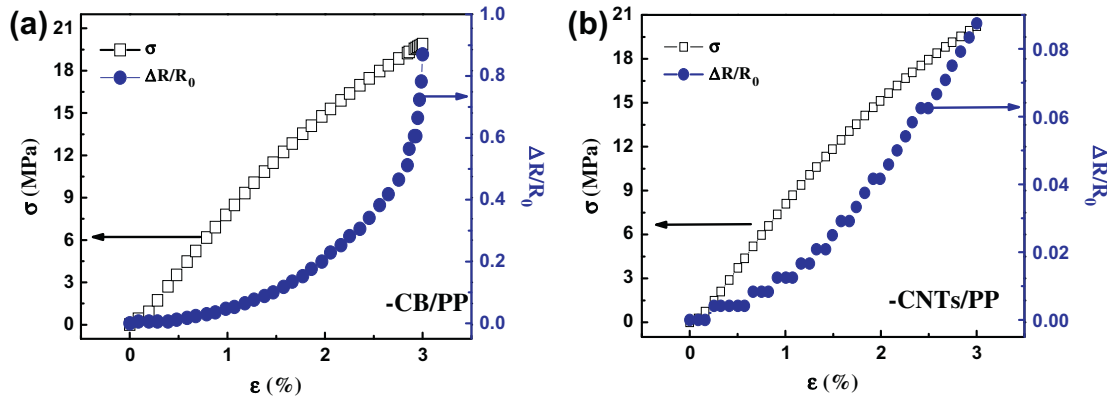


Fig. 7. Tensile stress (left axis) and normalized change of electrical resistivity (right axis) were plotted as a function of strain for composites: (a) CB (7.16 vol.%) / PP and (b) CNTs (3.41 vol.%) / PP. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of conductive paths formed by CB particles are nearly the same, leading to the very slight increase of $\Delta R/R_0$. With increasing the strain further, $\Delta R/R_0$ increases rapidly, this is because the formation and breakdown of conductive networks cannot balance each other easily, that is, the breakdown of conductive networks are more predominant than the formation of conductive networks in tension. Furthermore, it is found the $\Delta R/R_0$ increases exponentially with the strain, as depicted in Fig. 8a.

This trend of the curve of CB/PP can be explained by the tunneling mechanism proposed by Balberg [31]:

$$\sigma_{tun} \propto \exp \left[-\frac{r-2b}{d} \right] \quad (1)$$

where σ_{tun} is the inter-particle tunneling conductivity, r is the distance between the centers of the particles, b is the radius of the particles and d is the typical tunneling range (or tunneling decay) parameter. As clearly suggested by Eq. (1), every two particles (here assumed spherical) in the system are electrically connected, and their contribution to the conduction of the whole network diminishes as r increases.

Under the action of the tensile stress, the CB particles motion gradually. When the strain is small, the formation and breakdown of conductive paths constructed by contact of CB particles are balanced, resulting in a minute change of r , and a slow decrease of the

conductivity, i.e. the increase of $\Delta R/R_0$, was displayed. With increasing the strain further (higher than 0.5%), the distances between many CB particles grow markedly, thus largely dominating the resistivity of CB/PP. Considering the tunneling mechanism (as Eq. (1)), the raise of r results in obvious decrease of the conductivity, i.e. the increase of the resistivity. As a result, the resistivity of CB/PP increases significantly with increasing strain.

However, for CNTs/PP, at the beginning of the tension, several stages can be observed, this is because the large aspect ratio and entanglement structure of CNTs result in a balance of the formation and breakdown of conductive networks in perpendicular and parallel direction of tension. Subsequently, $\Delta R/R_0$ increases in a linear fashion with strain. It is believed the origin is the nice conductive network in CNTs/PP with a number of conductive contacts and stable network structure [20]. In addition, it is found CB/PP shows a higher $\Delta R/R_0$ than that of CNTs/PP with the same strain. The difference in the microstructure of conductive network, which leads to different responsive capability to the strain, is considered as the main reason. CB/PP has a more sensitive conductive network.

For the sake of evaluating the reproducibility of the strain sensor and get further information about the microstructure development of CPCs in tension, 10 cycles of the strain and $\Delta R/R_0$ vs. time of both CPCs was investigated. As observed from the plots of Fig. 8, the $\Delta R/R_0$ of both CPCs follows the strain, that is, it increases with

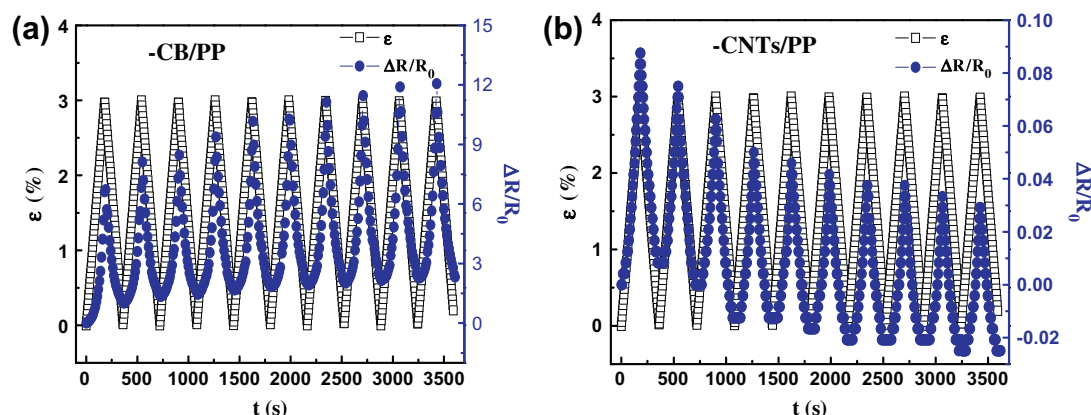


Fig. 8. Relationships of strain and $\Delta R/R_0$ vs. time of (a) CB (7.16 vol.%) /PP and (b) CNTs (3.41 vol.%) /PP composites during 10 extension–retraction cycles. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

The changes of max and min $\Delta R/R_0$ values of CB (7.16 vol.%) /PP and CNTs (3.41 vol.%) /PP during 10 cycles.

Cycle number		1	2	4	6	8	10
CB/PP	max $\Delta R/R_0$	6.72	8.12	9.39	10.28	11.46	12.06
	min $\Delta R/R_0$	1.08	1.34	1.71	1.98	2.24	2.48
CNTs/PP	max $\Delta R/R_0$	0.088	0.075	0.050	0.042	0.038	0.029
	min $\Delta R/R_0$	0.0083	0	−0.013	−0.020	−0.021	−0.025

the applied tensile; and decreases with the retraction of strain. This is because when the sample was dragged, the conductive particles will move with the polymer chains and pulled apart, resulting in an increase of interparticle distance, increasing relative resistance of samples [32]; while retraction, contrary to the stretch, the relative resistance of samples decreased.

More interestingly, for CB/PP, it is found the value of the max $\Delta R/R_0$ increases gradually with increasing the cycle number. The origin is discussed in detail as follows. Take the first extension–retraction cycle as an example (see Fig. 8a), when the strain is increased to the maximum 3%, the breakdown of conductive network plays a dominant role, CB particles motion with PP macromolecules, resulting in large distances between CB particles, i.e. a high $\Delta R/R_0$ value. In the subsequent retraction process, the distances between CB particles decrease gradually due to the releasing of the stress. However, at the end of this cycle, the position change of CB caused by the hysteresis effect resulting from matrix viscoelastic effects is hardly to be eliminated totally. Some vulnerable conductive paths formed by CB particles and aggregates with a low aspect ratio have been destroyed, and a notable increase of the min $\Delta R/R_0$ value to 1.08 in the first cycle is observed. In the extension process of the second cycle, the partially damaged conductive network will be broken further, distances between CB is further increased, leading to an obvious increase of max $\Delta R/R_0$ value (the value increases to 8.12 in the second cycle from 6.72 of the first cycle), so does the min $\Delta R/R_0$ value (it increases to 1.34 from 1.08 in the first cycle). With increasing the cycle number from 1 to 10, CB particles positions are arranged gradually in PP matrix, and increasing max and min $\Delta R/R_0$ values are shown (see Table 2, for simplification, only the first cycle and even number cycles are presented here). The max and min $\Delta R/R_0$ values in the tenth cycle are nearly two times higher than the values in the first cycle, respectively.

In contrast, for the CNTs/PP system, the max $\Delta R/R_0$ value becomes more and more negative during 10 cycles just as reported in literature [33]. The min $\Delta R/R_0$ value also becomes smaller and smaller (see Fig. 8b). It is known conductive network constructed by CNTs with large aspect ratio and entangled structure is very stable. A nice network is beneficial to avoiding the fracture of conduc-

tive path. Moreover, during cycles, the CNTs conductive network is influenced by hysteresis effect resulting from the viscoelastic behavior of CNTs/PP on the molecular level [20], the interaction between CNTs and the viscoelastic nature of the matrix leads to the rearrangement of CNTs and results in a better conductive network along the stretching direction after every cycle. A negative variation of the max and min $\Delta R/R_0$ value is therefore displayed (see Table 2). At the end, the max and min $\Delta R/R_0$ values in the tenth cycle become nearly three times lower than the values in the first cycle.

Based on the results and discussions, a schematic illustration is proposed to explain the mechanism of the different sensing behaviors between CB/PP and CNTs/PP as shown in Fig. 9. The original conductive networks before stretching is shown in Fig. 9a and a'. It can be seen that CNTs with large aspect ratio are arranged in matrix intricately; a nice conductive network by contact of CB particles is also achieved. For the state in tensile process with a high strain as shown in Fig. 9b and b', the breakdown of conductive networks are more predominant than the formation of new ones. For CNTs/PP, the number of conductive paths decreases obviously, while CNTs are rearranged in matrix along the tensile direction [34,35]. During the retraction process, some new conductive paths are formed, especially along the tensile direction. Owing to the hysteresis effect resulting from the viscoelastic behavior of the composites, some of these new conductive paths are preserved, and a much better conductive network is obtained at the end of this cycle (see Fig. 9c), leading to a decreasing max and min $\Delta R/R_0$. For CB/PP in Fig. 9b, due to the relatively weak interaction between the CB particles and aggregates (they have low aspect ratio), after this cycle, though some breakdown of conductive networks will reform again, the number of gaps between CB particles and the gaps' sizes both increase, leading to a increasing max and min $\Delta R/R_0$ as shown in Fig. 8a.

It is evident that the distinction in morphology and microstructure of the two fillers results in the obviously different strain sensing behaviors of the composites. The mechanism of the difference as discussed above of electrical conduction is meaningful for the long-term and cyclic applications of the strain sensors made from CPCs filled by different conductive fillers.

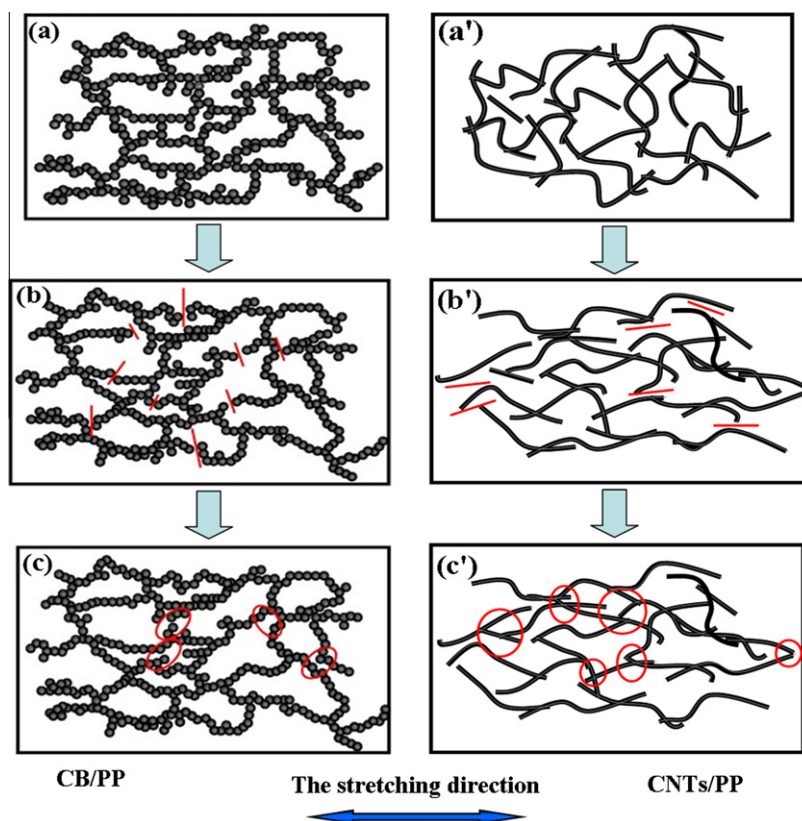


Fig. 9. Schematic illustration of influence of CB and CNTs structure on the strain sensing behavior of CB/PP (a–c) and CNTs/PP (a'–c'): (a and a') initial state; (b and b') the state in the tensile process with a high strain; (c and c') final state of filler distribution. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Conclusions

CB/PP and CNTs/PP composites with different filler concentration were prepared through the same processing. SEM observation showed that a nice dispersion of CB and CNTs in PP matrix was achieved. The percolation behaviors of the both CPCs were studied. A percolation threshold *ca.* 3.24 vol.% of CNTs/PP, which is much lower than that of CB/PP, is obtained.

The evolution about the stress and $\Delta R/R_0$ as a function of strain of the two CPCs was investigated. It was found for CB/PP, $\Delta R/R_0$ increases exponentially with the strain. This phenomenon was explained by tunneling mechanism. However, for CNTs/PP, $\Delta R/R_0$ increased in a linear fashion. It is proposed that the main origin of the different trend is the difference in the microscopic morphology of the two fillers, which leads to different conductive networks.

For the sake of evaluating the reproducibility of the strain sensor, the strain and $\Delta R/R_0$ vs. time for both CPCs during 10 cycles were investigated in detail. More interestingly, the values of the max and min $\Delta R/R_0$ in CB/PP and CNTs/PP showed obviously opposite trend with increasing the cycle number, i.e., the positive and the negative effect, indicating a different development of the conductive network. In the end of the tenth cycle, for CB/PP, the max and min $\Delta R/R_0$ values were nearly two times higher than the values in the first cycle; however, for CNTs/PP, the max and min $\Delta R/R_0$ values became nearly three times lower than the corresponding values in the first cycle.

It is suggested upon tension, for CNTs/PP, CNTs were rearranged in matrix along the tensile direction [34,35], the number of conductive paths on the parallel direction of tension increased based on the characteristics of CNTs with a large aspect ratio and the entangled structure. During extension–retraction cycles, a better conductive network was gradually formed due to the hysteresis

effect resulting from viscoelastic behavior of the composites; the decrease of max and min $\Delta R/R_0$ was thus obtained. However, for CB/PP, owing to the relatively weak interaction between the CB particles and aggregates, after several cycles, the number of gaps between CB particles and the gaps' sizes both increased, leading to an increasing max and min $\Delta R/R_0$. A schematic illustration was proposed to explain the mechanism of the different sensing behaviors between CB/PP and CNTs/PP.

The difference in the change of electrical property when two CPCs subjected to cyclical tension mainly depends on the distinction in the microstructure of conductive fillers, the interaction between fillers and matrix and the hysteresis effect resulting from the viscoelastic behavior of composites. These results are useful in designing CB and CNT-based polymer composites with different strain sensing capability to fulfill different requirements.

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